Open Source Accessibility

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Student Materials

Worksheet: - web - PDF

Group Discussion: - web - PDF

Overview

Ethics background required: It is helpful if students are familiar with the consequentialist and virtue ethics methods of considering ethical dilemmas. They should understand what stakeholders are. For any who need it, there are brief reviews of these concepts in the assignment. It is suggested that the Media Literacy Lab be completed prior to this lab, but principles of finding credible articles are reviewed in the student homework handout.

Subject matter referred to in this Lab: Linux or other open source software.

Placement in overall ethics curriculum:

- Academic year: Year 2-4 core course that uses open source software (OS works if students use Linux)
- Recommended previous labs: Foundational labs and media literacy lab
- Recommended follow-up labs: No direct follow-up lab

Time required:

- In class: 20-30 minutes;
- Out of class: 30 minutes

Learning objectives:

- Students apply what they know about media literacy to finding articles on open source software
- Understand the terms “open source” and “proprietary” and consider the benefits of each
- Expand student’s knowledge in the area of the ethical issue of accessibility

- Evaluate the accessibility of open source software using the consequentialist and virtue ethics frameworks
- Practice articulating and supporting a view about using open source software in light of accessibility issues
- Students have the opportunity to find a creative, ethical solution to creating accessible software

Ethical issues to be considered: Accessibility, Media Literacy

This lab requires students to complete some homework outside of class. They will be finding credible articles that consider the pros and cons of open source software and posting summaries online prior to class. During class they will be discussing within groups how open source and proprietary software measure up in regard to accessibility (economic, mental, and physical).

Preparation: Read the entire lab. Post the homework assignment online (or print on papers if you would like to have the assignment completed off-line. Print the group discussion questions either one per group or one per student.

Flow

- For homework, students find and read three articles from credible sources describing the pros and cons of open source software vs. proprietary. Results of this research should be posted online or submitted prior to the classwork part of this lab. Find student instructions here
- In class, the instructor introduces the idea of accessibility. The class brainstorms together on the different types of disabilities and what accommodations might be required in software products to address them.
- Students work in groups considering questions on the accessibility handout
- Students complete an exercise that shows that the virtue ethics framework doesn't clarify the accessible software dilemma, and that the consequentialist framework might be better.

Guide for Instructors

Lesson plan

Introduction (to be read or summarized to class) (5 min)

Ask students: “In your own words, what is open source software?” (definition is on handout)

One of the strongest arguments for open source software is that it is free, making it economically accessible to most people. In this discussion, we will consider other aspects of accessibility with regard to open source software. Accessibility is one of the current ethical considerations that software engineers must take into account as they design new software.

The Oxford Dictionary defines accessibility as: > "The quality of being able to be reached or entered."

It further explains accessibility as

1.1 The quality of being easy to obtain or use. 'students were concerned about the accessibility of quality academic counselling'.

1.2 The quality of being easily understood or appreciated. 'the accessibility of his work helped to popularize modern art'

1.3 The quality of being easily reached, entered, or used by people who have a disability. 'many architects believe that accommodating wheelchairs is all there is to providing accessibility'.

Have students brainstorm on the disabilities (learning, mental, emotional, physical) that people have to deal with today. Then ask them to come up with reasonable accommodations that software engineers might be able to include to help with these issues. The chart below provides the instructor with some ideas. There are many more.

Disability Software accommodation Blindness Audible output Color blindness Particular color schemes (don't use red and green or pink and gray together) Deafness Closed caption or other written instructions/descriptions Limited mobility Special hardware Dyslexia Choice of font (Arial and Courier are good) ADHD Good spacing in text – pages that are not too "busy" Chronic pain Interfaces that reduce the amount of movement required Depression Clear written instructions, allows for remote (work from home) communication Anxiety Simple interfaces and instructions Autism/Asperger's Good spacing in text – pages that are not too "busy"

Activity (10 min)

Students evaluate open source and proprietary software with regard to accessibility (10 min)

- Divide students into groups of 3-5 and pass out the student group discussion handouts. Students are asked to answer some questions based on their reading homework.
- If students are struggling, here are some potential answers to the questions:

- Is open source software easy to obtain and use?
  - It is easy to obtain in that it is free.
  - Sometimes the software downloading instructions might not be as clear as they are for proprietary software that has a whole team writing the manuals
  - Are there more bugs in open source software? Debatable. With more eyes on it there are more people to find and correct bugs. Proprietary software might have rushed deadlines so testing is not as thorough as desired. On the other hand proprietary software should have a testing team along with a documents team and a helpdesk.
  - Is open source easy to use? Also debatable. Again, no helpdesk for free versions, but lots of open forums.
  - A user would have to consider if the software would run on the user's current operating system, or if a virtual machine would be required?
  - Open source is not locked into a particular vendor's platform so it is often usable by more people.
  - Your students should have other thoughts from their reading.
- Is open source software easily understood?
  - Is the interface good?
  - Are there sufficient manuals? If the user wants to configure or change the code is the documentation (comments in code) good?
  - Since many people are touching the code, have they kept the documentation current? This is probably more likely to happen with proprietary software, but again, those open forums are invaluable!
- Is open source software easy to access by people with disabilities? Be sure to think about all of the different disabilities that you discussed with your class.

  *Disabilities can be physical (visual, auditory, motor), mental, or cognitive (learning disabilities).
  - Proprietary software designers may be more likely to include features like interfaces with attention to color-blindness, voice recognition, or audible readings of software.
  - Since open source software has more people looking at it, the diversity might allow for people who are sensitive to less common disabilities to make unique changes.
  - Open source was created to meet the needs of the original author, so might not, at least initially be very robust.

Reflection as a class

Ask students what the problem is with an ethical analysis using virtue ethics (both approaches – proprietary and open source violate various virtues). Ask students to suggest some of the virtues violated.

- Virtue ethics analysis (students may arrive at either answer using virtue ethics – the idea is to get them to think!) Some of the virtues that might be violated with proprietary are generosity and transparency. With open source, quality and hospitality (bad documentation) could be issues.
- In trying to come up with a consequentialist solution (helpful to the most people) students may come up with hybrid solutions such as the following:
  - Basic software is open source, but customers can pay for customized accessibility features.
  - Large companies support programmers in open source projects as add-ons to proprietary projects.
  - Software is free, but training and documentation cost.
  - Open source project display advertisements.

Assessment

Ask students if the discussion changed the way that they look at open source or proprietary software

Ask students what they learned about designing software with an eye for accessibility

Ask students to provide 2 accessibility issues with open source software

Ask students to provide 2 accessibility issues with proprietary software

Ask if virtue ethics or consequentialism was a better framework for considering this ethical dilemma and why

Bibliography